

TechMart Retail Management System

Business Requirements Document (BRD)

Version: 1.0

Prepared By: Database Design Team

Project: TechMart Retail Management System

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1. Project Overview

TechMart is a retail company specializing in the sale of consumer electronics, including smartphones, laptops, tablets, smartwatches, and accessories.

The company currently manages its operations using spreadsheets, which has resulted in several operational challenges such as:

- Data redundancy
- Difficulty generating reports
- Inaccurate inventory tracking
- Poor sales monitoring
- Manual order management

To improve operational efficiency, TechMart requires the development of a centralized Retail Management System.

2. Business Objectives

The proposed system should enable the company to:

- Manage customer information.
- Manage product catalog.
- Manage product categories.

- Manage suppliers.
- Manage employees.
- Record customer orders.
- Record products included in each order.
- Process customer payments.
- Monitor inventory levels.
- Generate business reports for management.

3. Functional Requirements

Customer Management

The system shall:

- Store customer information.
- Assign a unique identifier for every customer.
- Store customer name.
- Store phone number.
- Store email address.
- Store city and address.
- Allow one customer to place multiple orders.

Product Management

The system shall:

- Store product information.
- Assign a unique identifier for every product.
- Store product name.
- Store product description.
- Store selling price.
- Store available stock quantity.
- Associate each product with one category.

- Associate each product with one supplier.

Category Management

The system shall:

- Store product categories.
- Ensure each category has a unique name.
- Allow one category to contain multiple products.

Supplier Management

The system shall:

- Store supplier information.
- Assign a unique identifier for every supplier.
- Store company name.
- Store phone number.
- Store email address.
- Store city.
- Allow one supplier to provide multiple products.

Employee Management

The system shall:

- Store employee information.
- Assign a unique identifier.
- Store employee name.
- Store job title.
- Store hiring date.

- Store salary.
- Store branch information.

Order Management

The system shall:

- Record customer orders.
- Assign a unique order number.
- Store order date.
- Associate each order with one customer.
- Associate each order with the employee who created it.
- Store order status.

Order Details

The system shall:

- Allow one order to contain multiple products.
- Allow one product to appear in multiple orders.
- Store quantity for each ordered product.
- Store product selling price at the time of purchase.

Payment Management

The system shall:

- Record payment information.
- Associate one payment with one order.
- Store payment amount.
- Store payment date.

- Support multiple payment methods:
 - Cash
 - Credit Card
 - Vodafone Cash

4. Business Rules

- Every customer must have a unique Customer ID.
- Every category name must be unique.
- Every product belongs to exactly one category.
- Every product is supplied by exactly one supplier.
- Every order must belong to one customer.
- Every order must contain at least one product.
- Every payment belongs to one order.
- A customer cannot be deleted if they have existing orders.
- A category cannot be deleted while products are assigned to it.

5. Reports Required

Management requires the following reports:

Sales Reports

- Daily sales.
- Monthly sales.
- Best-selling products.
- Least-selling products.

Customer Reports

- Top purchasing customers.
- Inactive customers.

Product Reports

- Low-stock products.
- Average product prices.
- Product count by category.

Supplier Reports

- Number of products supplied by each supplier.

Employee Reports

- Number of orders handled by each employee.

6. Assumptions

- The system will operate only in Egypt.
- All prices are stored in Egyptian Pounds (EGP).
- One supplier supplies one or more products.
- Every order contains at least one product.
- Payment is recorded after order creation.

7. Project Scope

Included

- Database Design
- ERD
- Mapping
- SQL Server Implementation
- Data Analysis Queries
- Power BI Dashboard

Excluded

- Authentication System
- Mobile Application
- Website
- Online Payment Gateway

8. Deliverables

The project will include:

- Business Requirements Document
- System Analysis Document
- Entity Relationship Diagram (ERD)
- Database Mapping
- SQL Implementation
- Sample Dataset
- Analytical SQL Queries
- Power BI Dashboard